

Probabilistic Motion Models

Autonomous Navigation of Mobile Robots

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13.03.2025

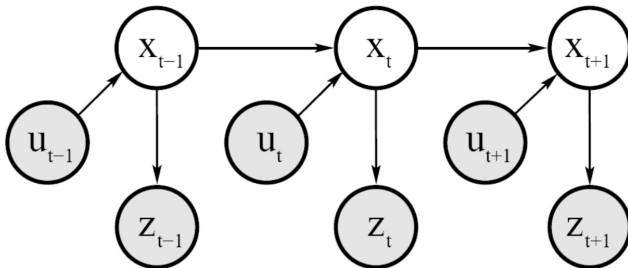
Robot Motion

- Robot motion is inherently uncertain
- How can we model this uncertainty?



Dynamic Bayesian Network

for Controls, States, and Sensations

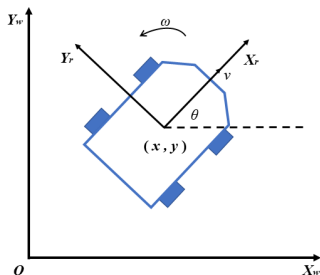


Probabilistic Motion Models

- To implement the Bayes Filter, we need the transition model $p(x_t | x_{t-1}, u_t)$
- The term $p(x_t | x_{t-1}, u_t)$ specifies a posterior probability, that action u_t carries the robot from x_{t-1} to x_t
- In this section we will discuss, how $p(x_t | x_{t-1}, u_t)$ can be modeled based on the motion equations and the uncertain outcome of the movements

Coordinate Systems

- The configuration of a typical wheeled robot in 3D can be described by six parameters
- This are the three-dimensional Cartesian coordinates plus the three Euler angles for roll, pitch, and yaw

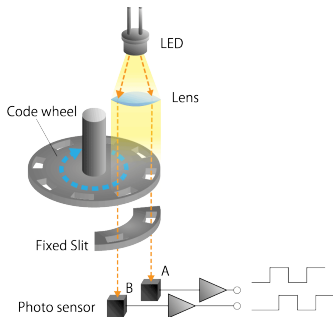


- For simplicity, throughout this section we consider robots operating on a planar surface
- The state space of such systems is three-dimensional (x, y, θ)

Typical Motion Models

- In practice, one often finds two types of motion models:
 - Odometry-based
 - Velocity-based (dead reckoning)
- Odometry-based models are used when systems are equipped with wheel encoders
- Velocity-based models have to be applied when no wheel encoders are given
- They calculate the new pose based on the velocities and the time elapsed

Wheel Encoders

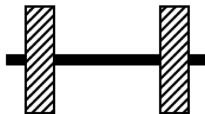


- The circuit module provide $\sim 5\text{ V}$ output when it “see” white, and a 0 V output when it “see” black
- Disks are manufactured out of high quality laminated color plastic to offer a very crisp black to white transition
- This enables a wheel encoder sensor to easily see the transitions

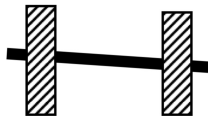
Dead Reckoning

- Derived from “deduced reckoning”
- Mathematical procedure for determining the present location of a vehicle
- Achieved by calculating the current pose of the vehicle based on its velocities and the time elapsed
- Historically used to log the position of ships

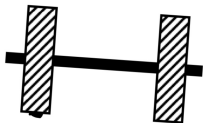
Reasons for Motion Errors of Wheeled Robots



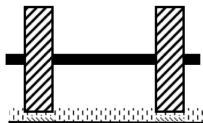
ideal case



different wheel diameters



bump



carpet

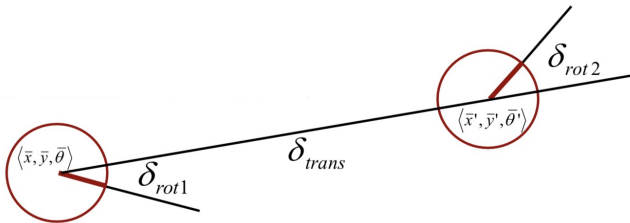
Odometry Model

- Robot moves from $\langle \bar{x}, \bar{y}, \bar{\theta} \rangle$ to $\langle \bar{x}', \bar{y}', \bar{\theta}' \rangle$
- Odometry information $u = \langle \delta_{rot1}, \delta_{rot2}, \delta_{trans} \rangle$

$$\delta_{trans} = \sqrt{(\bar{x}' - \bar{x})^2 + (\bar{y}' - \bar{y})^2}$$

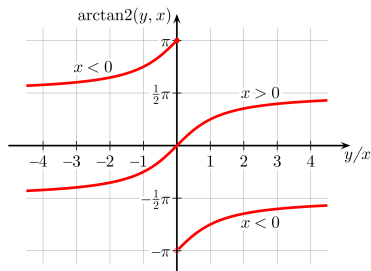
$$\delta_{rot1} = \text{atan2}(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta}$$

$$\delta_{rot2} = \bar{\theta}' - \bar{\theta} - \delta_{rot1}$$



The atan2 Function

$$\text{atan2}(y, x) = \begin{cases} \text{atan}(y/x) & \text{if } x > 0 \\ \text{sign}(y) (\pi - \text{atan}(|y/x|)) & \text{if } x < 0 \\ 0 & \text{if } x = y = 0 \\ \text{sign}(y) \pi/2 & \text{if } x = 0, y \neq 0 \end{cases}$$



Extends the inverse tangent and correctly copes with the signs of x and y

Noise Model for Odometry

The measured motion is given by the true motion corrupted with noise

$$\hat{\delta}_{rot_1} = \delta_{rot_1} + \varepsilon_{\alpha_1} |\delta_{rot_1}| + \alpha_2 |\delta_{trans}|$$

$$\hat{\delta}_{trans} = \delta_{trans} + \varepsilon_{\alpha_3} |\delta_{trans}| + \alpha_4 (|\delta_{rot_1}| + |\delta_{rot_2}|)$$

$$\hat{\delta}_{rot_2} = \delta_{rot_2} + \varepsilon_{\alpha_5} |\delta_{rot_2}| + \alpha_6 |\delta_{trans}|$$

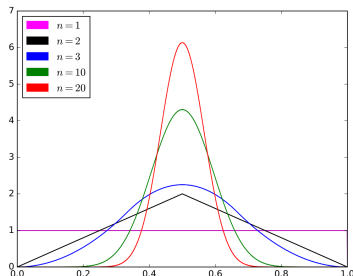
Typical Distributions for Probabilistic Motion Models

Normal distribution

$$\varepsilon_{\sigma^2}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left\{-\frac{1}{2} \frac{x^2}{\sigma^2}\right\}}$$

Triangular distribution

$$\varepsilon_{\sigma^2}(x) = \begin{cases} 0 & \text{if } |x| > \sqrt{6\sigma^2} \\ \frac{\sqrt{6\sigma^2} - |x|}{6\sigma^2} & \text{otherwise} \end{cases}$$



Calculating the Probability Density (zero-centered)

- For a normal distribution

Algorithm `prob_normal_distribution(a,b):`

Return $\frac{1}{\sqrt{2\pi} b^2} \exp \left\{ -\frac{1}{2} \frac{a^2}{b^2} \right\}$

- For a triangular distribution

Algorithm `prob_triangular_distribution(a,b):`

Return $\max \left\{ 0, \frac{1}{\sqrt{6} b} - \frac{|a|}{6 b^2} \right\}$

Calculating the Posterior Given x, x' , and Odometry

Algorithm `motion_model_odometry(x, x',  $\bar{x}$ ,  $\bar{x}'$ )`:

$$\delta_{trans} = \sqrt{(\bar{x}' - \bar{x})^2 + (\bar{y}' - \bar{y})^2}$$

$$\delta_{rot_1} = \text{atan2}(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta}$$

$$\delta_{rot_2} = \bar{\theta}' - \bar{\theta} - \delta_{rot_1}$$

odometry
params (u)

$$\hat{\delta}_{trans} = \sqrt{(x' - x)^2 + (y' - y)^2}$$

$$\hat{\delta}_{rot_1} = \text{atan2}(y' - y, x' - x) - \theta$$

$$\hat{\delta}_{rot_2} = \theta' - \theta - \hat{\delta}_{rot_1}$$

values of
interest
(x, x')

$$p_1 = \text{prob}(\delta_{rot_1} - \hat{\delta}_{rot_1}, \alpha_1 |\delta_{rot_1}| + \alpha_2 \delta_{trans})$$

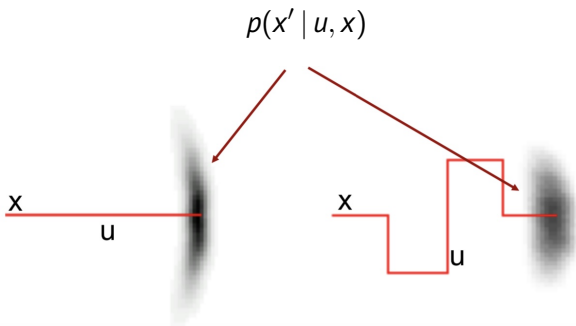
$$p_2 = \text{prob}(\delta_{trans} - \hat{\delta}_{trans}, \alpha_3 \delta_{trans} + \alpha_4 (|\delta_{rot_1}| + |\delta_{rot_2}|))$$

$$p_3 = \text{prob}(\delta_{rot_2} - \hat{\delta}_{rot_2}, \alpha_1 |\delta_{rot_2}| + \alpha_2 \delta_{trans})$$

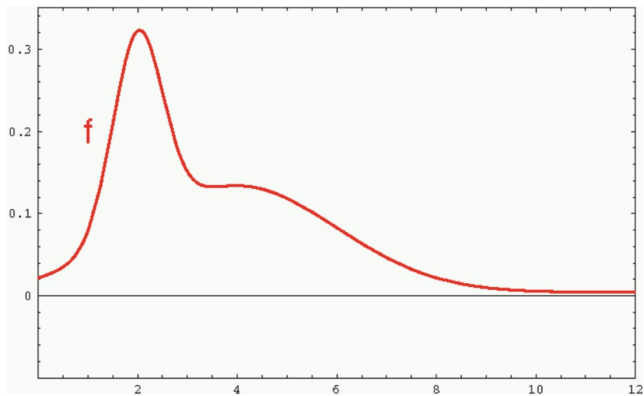
Return $p_1 \cdot p_2 \cdot p_3$

Application

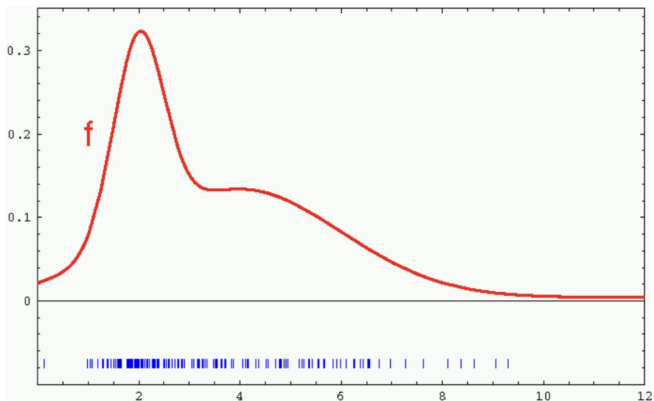
- Repeated application of the motion model for short movements
- Typical banana-shaped distributions obtained for the 2D-projection of the 3D posterior



Sample-Based Density Representation



Sample-Based Density Representation



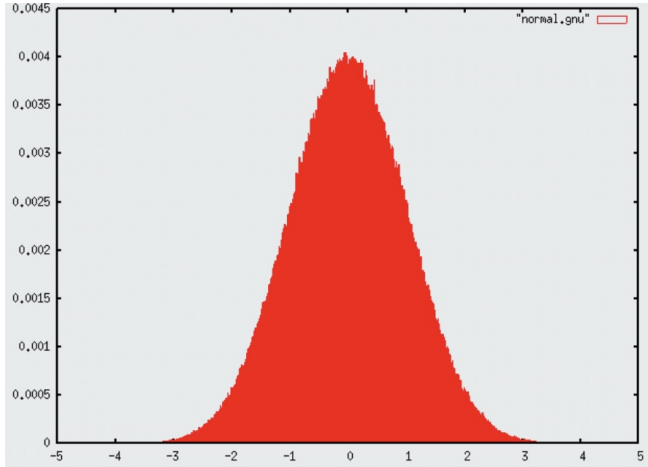
Sampling

- Sampling from a normal distribution

Algorithm `sample_normal_distribution(b)`:

Return $1/2 \sum_{i=1}^{12} \text{rand}(-b, b)$

Normally Distributed Samples



10^6 samples

Sampling

- Sampling from a normal distribution

Algorithm `sample_normal_distribution(b)`:

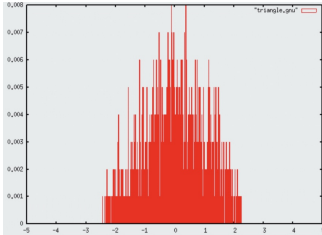
Return $1/2 \sum_{i=1}^{12} \text{rand}(-b, b)$

- Sampling from a triangular distribution

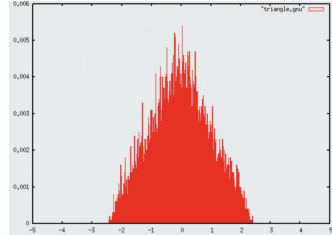
Algorithm `sample_triangular_distribution(b)`:

Return $\sqrt{6}/2 [\text{rand}(-b, b) + \text{rand}(-b, b)]$

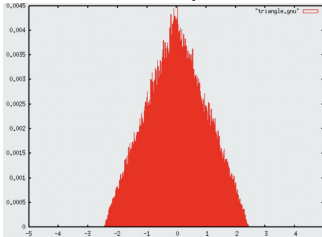
For Triangular Distribution



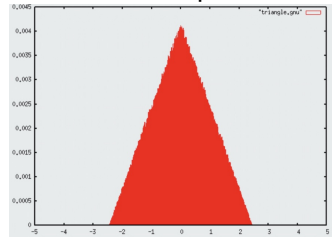
10^3 samples



10^4 samples

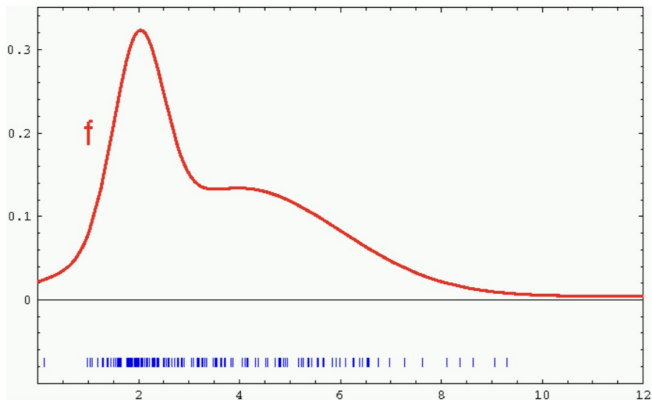


10^5 samples



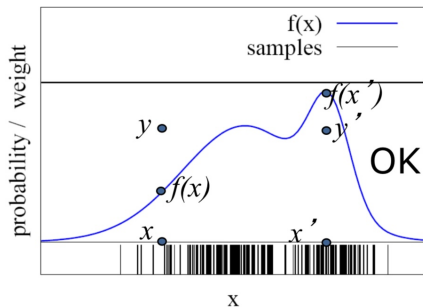
10^6 samples

How to Obtain Samples from Arbitrary Functions?



Rejection Sampling

- Sampling from arbitrary distributions
- Sample x from a uniform distribution from $[-b, b]$
- Sample y from $[0, \max f]$
- if $f(x) > y$ keep the sample x , otherwise reject it



Rejection Sampling

Sampling from arbitrary distributions

Algorithm `sample_distribution(f, b):`

Repeat

`x = rand(-b, b)`

`y = rand(0, max{f(x) | x ∈ [-b, b]})`

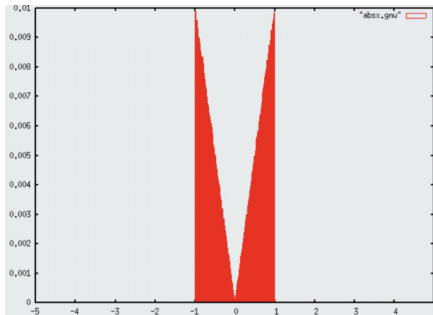
Until `y ≤ f(x)`

Return `x`

Example

Sampling from

$$f(x) = \begin{cases} \text{abs}(x) & x \in [-1, 1] \\ 0 & \text{otherwise} \end{cases}$$



Sample Odometry Motion Model

Algorithm `sample_motion_model(u, x)`:

$$u = \langle \delta_{rot_1}, \delta_{rot_2}, \delta_{trans} \rangle,$$

$$x = \langle x, y, \theta \rangle$$

$$\hat{\delta}_{rot_1} = \delta_{rot_1} + \text{sample}(\alpha_1 |\delta_{rot_1}| + \alpha_2 \delta_{trans})$$

$$\hat{\delta}_{rot_2} = \delta_{rot_2} + \text{sample}(\alpha_1 |\delta_{rot_2}| + \alpha_2 \delta_{trans})$$

$$\hat{\delta}_{trans} = \delta_{trans} + \text{sample}(\alpha_3 \delta_{trans} + \alpha_4 (|\delta_{rot_1}| + |\delta_{rot_2}|))$$

sample_
normal_
distribution

$$x' = x + \hat{\delta}_{trans} \cos(\theta + \hat{\delta}_{rot_1})$$

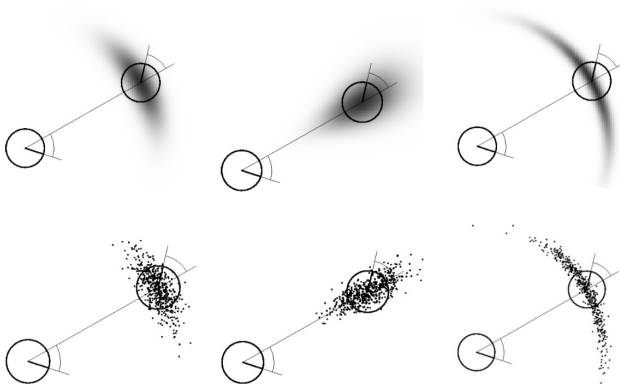
$$y' = y + \hat{\delta}_{trans} \sin(\theta + \hat{\delta}_{rot_1})$$

$$\theta' = \theta + \hat{\delta}_{rot_1} + \hat{\delta}_{rot_2}$$

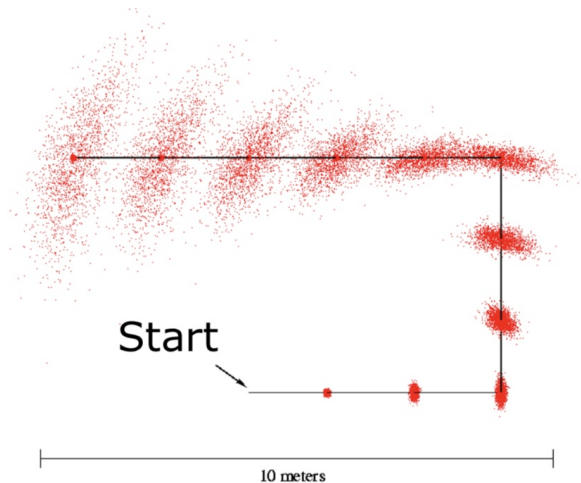
Return $\langle x', y', \theta' \rangle$

Examples

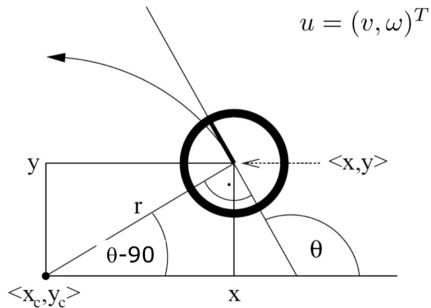
Odometry-Based



Sampling from Our Motion Model



Velocity-Based Model



Velocity-Based Model

Noise Model

- The measured motion is given by the true motion corrupted with noise

$$\hat{v} = v + \varepsilon_{\alpha_1 |v| + \alpha_2 |\omega|}$$

$$\hat{\omega} = \omega + \varepsilon_{\alpha_3 |v| + \alpha_4 |\omega|}$$

- What is the disadvantage of this noise model?

Velocity-Based Model

Noise Model

- The $(\hat{v}, \hat{\omega})$ -circle constrains the final orientation (2D manifold in a 3D space)
- Better approach

$$\hat{v} = v + \varepsilon_{\alpha_1} |v| + \alpha_2 |\omega|$$

$$\hat{\omega} = \omega + \varepsilon_{\alpha_3} |v| + \alpha_4 |\omega|$$

$$\hat{\gamma} = \varepsilon_{\alpha_1} |v| + \alpha_2 |\omega|$$

$\hat{\gamma}$ – to account for the final rotation

Motion Including 3rd Parameter

$$x' = x - \frac{\hat{v}}{\hat{\omega}} \sin \theta + \frac{\hat{v}}{\hat{\omega}} \sin(\theta + \hat{\omega} \Delta t)$$

$$y' = y + \frac{\hat{v}}{\hat{\omega}} \cos \theta - \frac{\hat{v}}{\hat{\omega}} \cos(\theta + \hat{\omega} \Delta t)$$

$$\theta' = \theta + \hat{\omega} \Delta t + \hat{\gamma} \Delta t$$

$\hat{\gamma}$ – to account for the final rotation

Equation for the Velocity Model

$$x_{t-1} = (x, y, \theta)^T$$

$$x_t = (x', y', \theta')^T$$

- Center of circle

$$\begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -\lambda \sin \theta \\ \lambda \cos \theta \end{pmatrix}$$

λ – some constant (distance to ICC)

Equation for the Velocity Model

$$x_{t-1} = (x, y, \theta)^T$$

$$x_t = (x', y', \theta')^T$$

■ Center of circle

$$\begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -\lambda \sin \theta \\ \lambda \cos \theta \end{pmatrix} = \begin{pmatrix} \frac{x + x'}{2} + \mu(y - y') \\ \frac{y + y'}{2} + \mu(x' - x) \end{pmatrix}$$

λ – some constant (distance to ICC)

μ – some constant (the center of the circle lies on a ray half way between x and x' and is orthogonal to the line between x and x')

Equation for the Velocity Model

$$x_{t-1} = (x, y, \theta)^T$$

$$x_t = (x', y', \theta')^T$$

- Center of circle

$$\begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -\lambda \sin \theta \\ \lambda \cos \theta \end{pmatrix} = \begin{pmatrix} \frac{x + x'}{2} + \mu(y - y') \\ \frac{y + y'}{2} + \mu(x' - x) \end{pmatrix}$$

- Allows us to solve the equations to

$$\mu = \frac{1}{2} \frac{(x - x') \cos \theta + (y - y') \sin \theta}{(y - y') \cos \theta - (x - x') \sin \theta}$$

Equation for the Velocity Model

$$x_{t-1} = (x, y, \theta)^T$$

$$x_t = (x', y', \theta')^T$$

$$\begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} \frac{x + x'}{2} + \mu(y - y') \\ \frac{y + y'}{2} + \mu(x' - x) \end{pmatrix} \quad \mu = \frac{1}{2} \frac{(x - x') \cos \theta + (y - y') \sin \theta}{(y - y') \cos \theta - (x - x') \sin \theta}$$

■ and

$$r^* = \sqrt{(x - x^*)^2 + (y - y^*)^2}$$

$$\Delta\theta = \text{atan2}(y' - y^*, x' - x^*) - \text{atan2}(y - y^*, x - x^*)$$

Equation for the Velocity Model

- The parameters of the circle

$$r^* = \sqrt{(x - x^*)^2 + (y - y^*)^2}$$

$$\Delta\theta = \text{atan2}(y' - y^*, x' - x^*) - \text{atan2}(y - y^*, x - x^*)$$

- allow for computing the velocities as

$$v = \frac{\Delta\theta}{\Delta t} r^* \quad \omega = \frac{\Delta\theta}{\Delta t}$$

Posterior Probability for Velocity Model

Algorithm `motion_model_velocity`(x_t, u_t, x_{t-1}): $p(x_t | x_{t-1}, u_t)$

$$\mu = \frac{1}{2} \frac{(x - x') \cos \theta + (y - y') \sin \theta}{(y - y') \cos \theta - (x - x') \sin \theta}$$

$$x^* = (x + x')/2 + \mu (y - y')$$

$$y^* = (y + y')/2 + \mu (x' - x)$$

$$r^* = \sqrt{(x - x^*)^2 + (y - y^*)^2}$$

$$\Delta t = \text{atan2}(y' - y^*, x' - x^*) - \text{atan2}(y - y^*, x - x^*)$$

$$\hat{v} = \Delta \theta / \Delta t \cdot r^*$$

$$\hat{\omega} = \Delta \theta / \Delta t$$

$$\hat{\gamma} = (\theta' - \theta) / \Delta t - \hat{\omega}$$

Return $\text{prob}(v - \hat{v}, \alpha_1 v^2 + \alpha_2 \omega^2) \cdot \text{prob}(\omega - \hat{\omega}, \alpha_3 v^2 + \alpha_4 \omega^2) \cdot$
 $\text{prob}(\hat{\gamma}, \alpha_5 v^2 + \alpha_6 \omega^2)$

Sampling

from Velocity Model

Algorithm `sample_motion_model_velocity( $u_t, x_{t-1}$ )`:

$$\hat{v} = v + \text{sample}(\alpha_1 v^2 + \alpha_1 \omega^2)$$

$$\hat{\omega} = \omega + \text{sample}(\alpha_3 v^2 + \alpha_3 \omega^2)$$

$$\hat{\gamma} = \text{sample}(\alpha_5 v^2 + \alpha_6 \omega^2)$$

$$x' = x - \frac{\hat{v}}{\hat{\omega}} \sin \theta + \frac{\hat{v}}{\hat{\omega}} \sin(\theta + \hat{\omega} \Delta t)$$

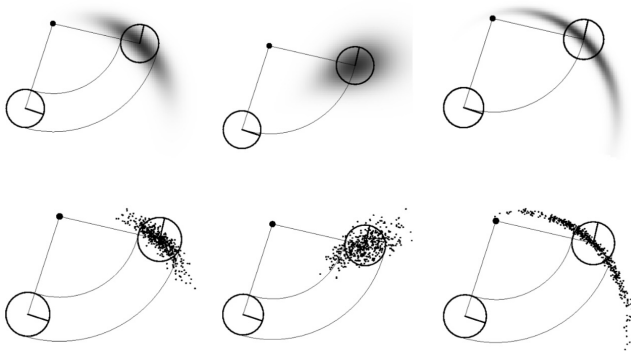
$$y' = y + \frac{\hat{v}}{\hat{\omega}} \cos \theta - \frac{\hat{v}}{\hat{\omega}} \cos(\theta + \hat{\omega} \Delta t)$$

$$\theta' = \theta + \hat{\omega} \Delta t + \hat{\gamma} \Delta t$$

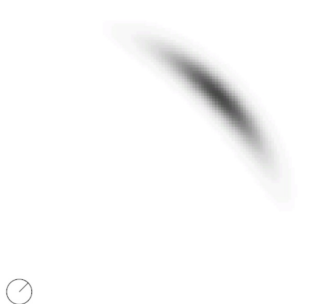
Return $(x', y', \theta')^T$

Examples

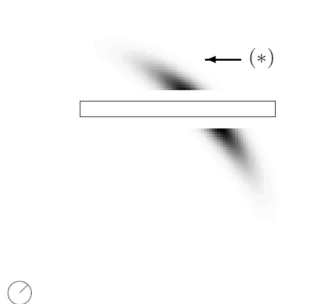
Velocity-Based



Map-Consistent Motion Model



$$p(x' | u, x)$$



$$p(x' | u, x, m)$$

Approximation: $p(x' | u, x, m) = \eta p(x' | m) p(x' | u, x)$